

2017-2018 – DEEQA – TSE – Advanced Macroeconomics

Lecture 6: News, Recessions and Liquidations

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Roadmap

1. Cycles Without Inflation
2. The News View
3. Liquidations
4. Economic Cycle: Some Further Evidence

Part I : Cycles Without Inflation

Roadmap of Part I

1. Motivations
2. A New Keynesian Model with Gains From Trade

Roadmap

1. Motivations
2. A New Keynesian Model with Gains From Trade

I.1. Motivations

The modern approach to business cycles fluctuations: Shocks

- ▶ The economy is hit by “shocks”,
- ▶ Realistic shocks are either “supply” or “demand”,
- ▶ Supply:
 - × Technology,
 - × Oil price,
 - × Taxes.
- ▶ Demand:
 - × Monetary shocks,
 - × Fiscal,
 - × Investors mood.
- ▶ Empirical work (Smets and Wouters) brings a lot of shocks (not always realistic) (preference shocks, markup shocks, shocks to arbitrage equations, etc...).

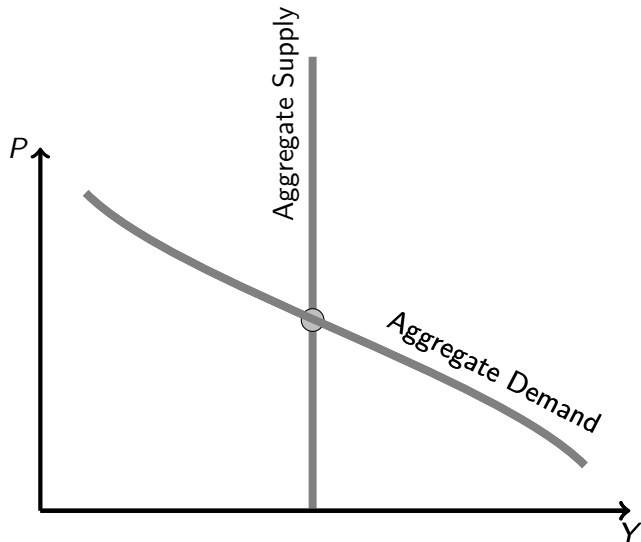
I.1. Motivations

The modern approach to business cycles fluctuations: Models

- ▶ Models are of two types: “Real Business Cycles” Models and “New-Keynesian” ones:
- ▶ Real Business Cycles:
 - × Flexibles Prices,
 - × Supply shocks,

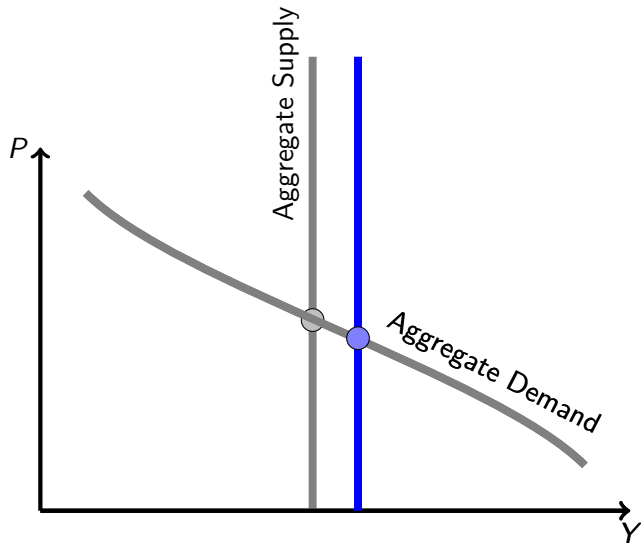
I.1. Motivations

Real Business Cycles Models



I.1. Motivations

Real Business Cycles Models



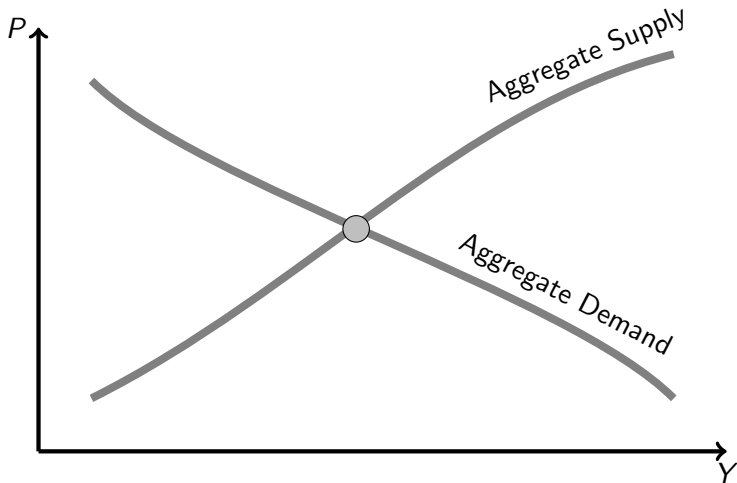
I.1. Motivations

The modern approach to business cycles fluctuations: Models

- ▶ Models are of two types: “Real Business Cycles Models” and “New-Keynesian” ones:
- ▶ New-Keynesian Models:
 - × Prices are sticky,
 - × Monetary rules (Taylor rules) matter,
 - × Demand shocks.

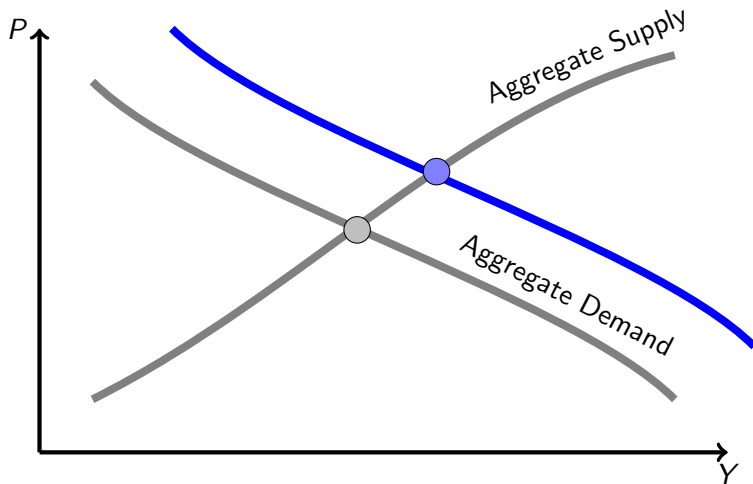
I.1. Motivations

New Keynesian Models



I.1. Motivations

New Keynesian Models



I.1. Motivations

The modern approach to business cycles fluctuations: Models

- ▶ Both models and shocks have hard time to explain the recent periods (last 30 years).

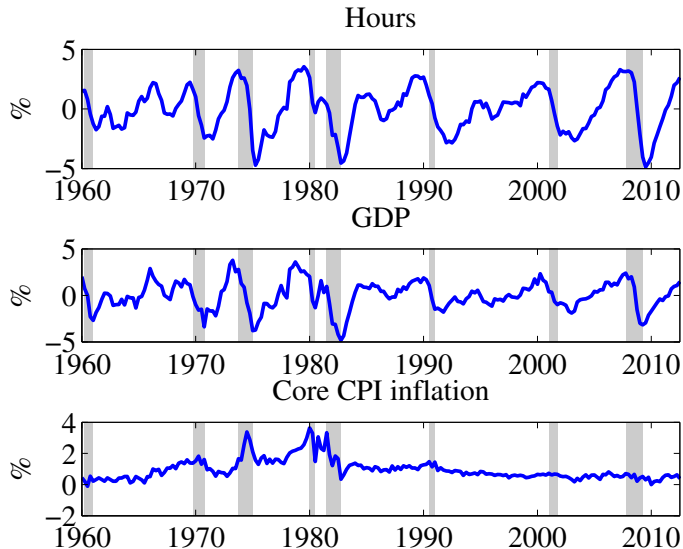
I.1. Motivations

Some Intriguing Facts over the last 3 cycles

- ▶ Over the last 30, the US economy (can be extended to the Eurozone) has experienced an intriguing type of business cycle.
- ▶ Let's look at
 - × Detrended hours H
 - × Detrended Y
 - × Raw core inflation

I.1. Motivations

Some Intriguing Facts over the last 3 cycles: Non inflationary business cycles



I.1. Motivations

Intriguing Facts for Usual Shocks and Models

- ▶ I discuss the following points:
- ▶ Demand shocks?
 - × Should be inflationary if they move output.
 - × Should not move output if not inflationary.
 - × If flex prices, C and I move in opposite direction.
- ▶ Supply shocks?
 - × Productivity should be procyclical
 - × Investment Specific Technology shocks: investment price should be countercyclical

I.1. Motivations

The Trouble with NK Models

- ▶ If output moves because the output gap moves (demand shocks), then we should expect inflation response.
- ▶ The core NK model is quantitatively off target.
- ▶ Take the Jordi Galí's textbook New Phillips curve

$$\pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t + u_t$$

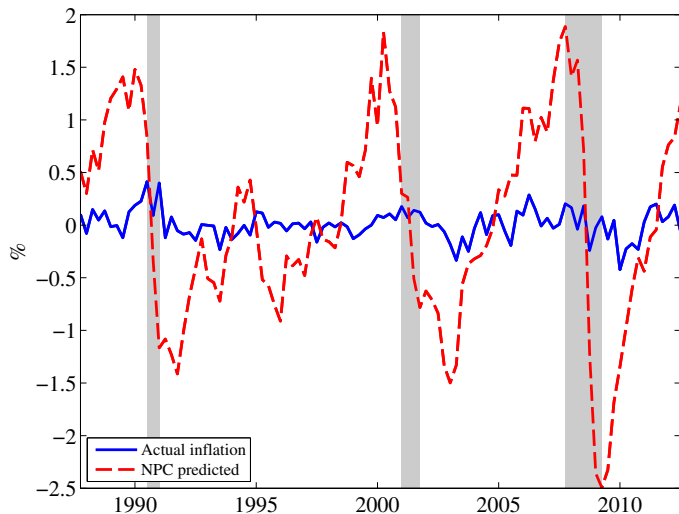
- ▶ Assume that the output gap is AR(1) with persistence ρ .

$$\pi_t = \frac{\kappa}{1 - \beta\rho} \tilde{y}_t + u_t$$

- ▶ Take Jordi's textbook calibration (including a mean duration of prices of 3 quarters).
- ▶ Assume that the output gap is measured by the HP cycle of output.
- ▶ Feed it into this last equation and deduct the implied inflation, killing cost-push shocks.

I.1. Motivations

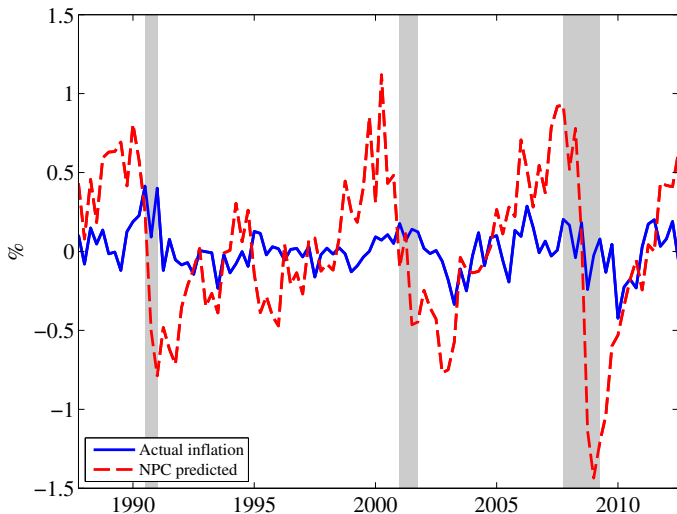
The Trouble with NK Models



- Post Volcker, NPC implies that s.d. of inflation is **350%** of the actual one

(Long) Introduction

The Trouble with NK Models - Output gap is AR(2)



- Post Volcker, NPC implies that s.d. of inflation is **177%** of the actual one

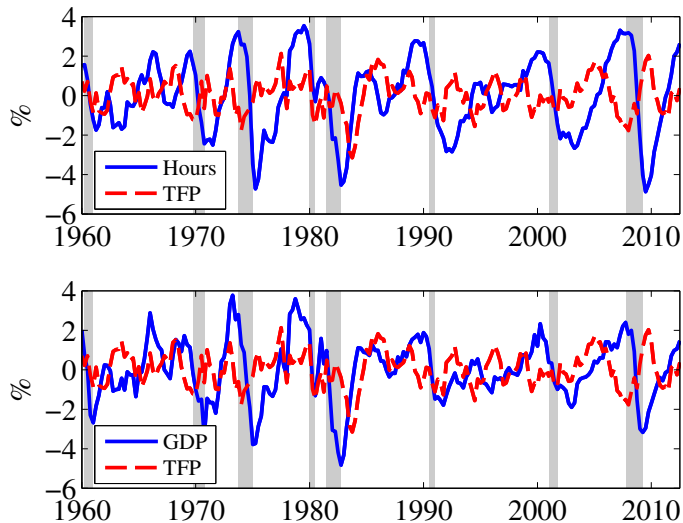
I.1. Motivations

The Trouble with RBC Models

- ▶ This pattern is hard to explain in the RBC tradition :
- ▶ No procyclical TFP shocks.
- ▶ No pro cyclical Investment Specific Technology shocks: investment relative prices *are not* countercyclical.
- ▶ Demand shocks typically cause negative consumption - investment co-movements.

I.1. Motivations

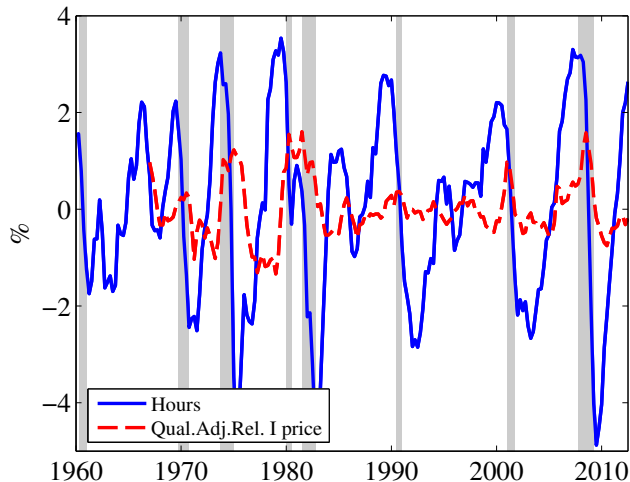
The Trouble with RBC Models: TFP



- Post-Volcker, correlation between hours worked and TFP is **-0.64**, correlation

I.1. Motivations

The Trouble with RBC Models: IST shocks



- Post-Volcker, correlation between hours worked and relative price of investment is **.56**.

I.1. Motivations

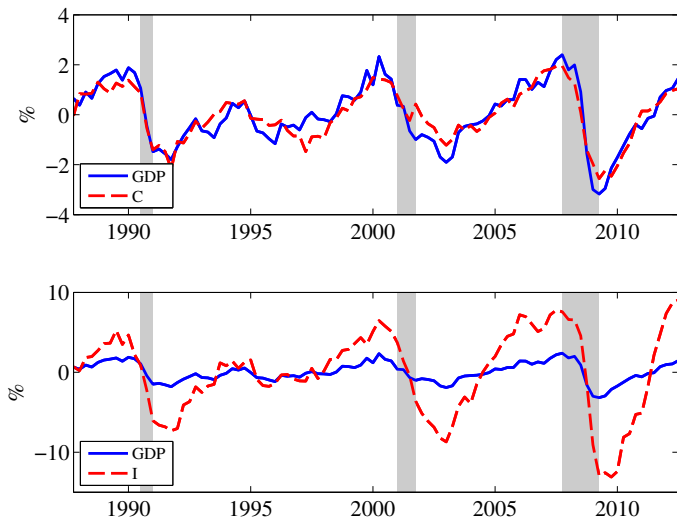
The Trouble with RBC Models: IST shocks

Table 1: Various measures of the relative price of investment, deflating with core CPI, correlations with Hours

Variable	1960Q1-2012Q3	Post-Volcker
Qual.Adj.I	-0.07	0.56
Fixed I	0.42	0.76
Non Res.I	0.09	0.63
Struct.I	0.44	0.75
Equip.I	-0.25	0.17
PPI Equip.	-0.24	0.11
Resid.I	0.70	0.80
SP500	0.31	0.56

I.1. Motivations

The Trouble with RBC Models: Demand Shocks



- Post-Volcker, correlations with HP filtered output are **.92** for C and **.91** for I .

I.1. Motivations

The Trouble with RBC and NK Models

- ▶ Possible to "fix" these commonly used RBC or NK models to fit facts: "Marginal Efficiency of Investment" shocks, preference shocks, fixed price ("backward-looking" Phillips curve), adjustment costs to the investment rate, in-sample correlation of shocks, etc...
- ▶ Those explanations in our opinion are not very compelling, intuitive or robust.
- ▶ I will review some. models that propose different mechanisms to interpret the data.
- ▶ One is related to the role of trade between agents.

I.1. Motivations

A Story

- ▶ Spain
- ▶ Two types of households
- ▶ Carpenters and Farmers
- ▶ Houses (capital good) and tomatoes (consumption good)
- ▶ In the short run, specialization is given.

I.1. Motivations

A Story (continued)

- ▶ The carpenter needs to eat, the farmer needs a shelter
- ▶ Static Gains from Exchange (from *Trade*) between the two.
- ▶ Assume that the perceived value of houses decreases (downward revision of expectations, bad news, pessimism, ...)
- ▶ The relative price of houses in terms of tomatoes p will go down
- ▶ The carpenter will work less for two reasons
 1. he does not want as many houses as before
 2. he cannot trade as many houses as before
- ▶ L_I and $I \searrow$
- ▶ The farmer does not want to buy as many houses as before, and does not need to produce as many tomatoes for the trade: L_C and $C \searrow$

I.1. Motivations

A Story (continued)

- ▶ $C \searrow, I \searrow, L \searrow$ in both sectors, $Y = C + pI \searrow$
- ▶ If reallocation of workers take some time, the recession is likely to be protracted.
- ▶ Changes in perceptions about the future are affecting the width of Gains from Trade today
- ▶ Fluctuations are here related with variations in the amount of Gains from Trade between agents.
- ▶ This is about *natural output* fluctuations $\rightarrow$ does not move inflation.

I.1. Motivations

To Be Compared To The Traditional Story

- ▶ The same household produces *both* houses and tomatoes
- ▶ If the perceived value of houses goes down, it is time to
 - × work less in constuction
 - × work more in farming
- ▶ $C \nearrow, I \searrow$

Roadmap of Part I

1. Motivations
2. A New Keynesian Model with Gains From Trade

I.2. A New Keynesian Model with Gains From Trade

The basic model

- ▶ Let's start with the textbook New Keynesian model.
- ▶ $c_t = \left(\int_0^1 c_{jt}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}$
- ▶ $\sum \beta^t (\ln(c_{ct}) + \Phi(1 - \ell_{ct}))$.
- ▶ Monopoly $j : C_{jt} = A_t L_{jt}$
- ▶ Calvo price setting
- ▶ Monetary authorities follow a Taylor rule.

I.2. A New Keynesian Model with Gains From Trade

The basic model

- ▶ This basic model collapses to a Phillips curve and a dynamic IS equation + Taylor rule

$$\left\{ \begin{array}{lcl} \tilde{y}_t & = & -(\hat{i}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t & = & \beta E_t \hat{\pi}_{t+1} + \lambda (\hat{y}_t - \hat{y}_t^n) \\ + & & \text{Taylor rule} \end{array} \right.$$

- ▶ Output gap: $\tilde{y}_t = \hat{y}_t - \hat{y}_t^n$
- ▶ $\hat{y}_t^n = \hat{A}_t$

I.2. A New Keynesian Model with Gains From Trade

The basic model

$$\left\{ \begin{array}{lcl} \tilde{y}_t & = & -(\hat{v}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t & = & \beta E_t \hat{\pi}_{t+1} + \lambda (\hat{y}_t - \hat{y}_t^n) \\ + & & \text{Taylor rule} \end{array} \right.$$

- ▶ What happens in this environment if agents expect $\hat{A}_{t+1}$ to be high?
- ▶ It is a demand shock to current consumption.
- ▶ Absent of an active Taylor Rule, inflation.
- ▶ The increased expectation of A_{t+1} does not directly enter into the Phillips curve.
- ▶ Therefore, if it creates an output boom, it will create inflation (no change in the natural output)

I.2. A New Keynesian Model with Gains From Trade

The basic model

$$\left\{ \begin{array}{lcl} \tilde{y}_t & = & -(\hat{v}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t & = & \beta E_t \hat{\pi}_{t+1} + \lambda (\hat{y}_t - \hat{y}_t^n) \\ + & & \text{Taylor rule} \end{array} \right.$$

- ▶ What happens in this environment if agents observe high $\hat{A}_t$?
- ▶ It is a supply shock .
- ▶ It shows up directly in the the Phillips curve ($\hat{y}_t^n = \hat{A}_t$).
- ▶ Therefore, if it creates an output boom, it will not create inflation (changes in the natural output)

I.2. A New Keynesian Model with Gains From Trade

The basic model

- ▶ Textbook prescription: to keep stable inflation, monetary authorities need to strongly counteract demand shock but need to accommodate supply shocks.
- ▶ If the economy goes into recession due to a fall in demand – as opposed to a reduction in supply capacity – this should put substantial downward pressure on prices.
- ▶ Let's contrast those results with the one of a gains-from-trade model.

I.2. A New Keynesian Model with Gains From Trade

The gains-from-trade model

- ▶ Add a mass 1 of agents (type 2)
- ▶ Type 2 individuals produce some capital good (full depreciation): $K_{t+1} = L_{2t}$.
- ▶ Preferences: $\ln \left(C_{2t} + \frac{\psi_2}{2}(1 - L_{2t})^2 \right)$.
- ▶ Capital market is flex-price and competitive.
- ▶ Production of consumption good: $C_{jt} = \Theta_t K_{jt} + L_{1jt}$
- ▶ If then mass of type 2 individuals is 0, then there is no capital supply and the model is the previous one.

I.2. A New Keynesian Model with Gains From Trade

The gains-from-trade model

- The log linear approximation of the model is

$$\left\{ \begin{array}{lcl} \tilde{y}_t & = & -\zeta (\hat{i}_t - E_t \hat{\pi}_{t+1} - \hat{\rho}_t^n) + E_t \tilde{y}_{t+1} \\ \hat{\pi}_t & = & \beta E_t \hat{\pi}_{t+1} + \lambda \zeta^{-1} (\hat{y}_t - \hat{y}_t^n) \\ + & & \text{Taylor rule} \end{array} \right.$$

- Natural or non-inflationary level of output $\hat{y}_t^n = \phi_2 A_t + \phi_1 E_t [\hat{\Omega}_{t+1}]$ where $\hat{\Omega}_{t+1} = \hat{\Theta}_{t+1} - \hat{A}_{t+1}$

I.2. A New Keynesian Model with Gains From Trade

A new new Phillips curve

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \lambda \zeta^{-1} \left(\hat{y}_t - \phi_2 A_t - \phi_1 E_t [\hat{\Omega}_{t+1}] \right)$$

- ▶ Consider believes that Ω_{t+1} will be high.
- ▶ Agents of type 1 to want to by capital as it return is expected to be high.
- ▶ They will trade more with agents of type 2.
- ▶ Type 2 agents will also want more capital, but also more consumption because of an income effect.
- ▶ This would look like a demand shock.
- ▶ But it does not put any pressures on prices, as the natural or non-inflation rate of output has also changed.
- ▶ Same thing with downward revisions: recession but no deflation.

I.2. A New Keynesian Model with Gains From Trade

A useful framework

- ▶ Useful model to think of the last episodes of non-inflationary expansions and recessions.
- ▶ Booms and busts are mainly driven by expectations on investment return.

Part II : The News View

Roadmap of Part II

1. The News View
2. Informal and VAR Evidence
3. An Analytical Framework
4. Discussion / Extensions
5. Applications

Roadmap of Part II

1. The News View
2. Informal and VAR Evidence
3. An Analytical Framework
4. Discussion / Extensions
5. Applications

II.1. The New view

- ▶ Here I present a set of facts/ideas that I think might be relevant
 - × to understand the three recent business cycles in the developed world (US, Europe)
 - × to understand some previous episodes like the 1997 asian crisis
 - × to design policies when looking forward at emerging economies
- ▶ It could be an interesting framework to discuss/work on issues like housing bubbles, waves of investment craze driven by expectations, etc...

II.1. The New view

The news view of business cycles

- ▶ Boom/bust are mainly the result of agents having incentives to continuously anticipate the economy's future demands.
- ▶ Properly anticipating a future need $\rightsquigarrow$, gains by trying to preempt the market and invest early as to make goods readily available when the predicted needs eventually appear.
- ▶ If many agents adopt similar behavior, because they receive related news about future developments, $\rightsquigarrow$ a boom period.
- ▶ However, by the very fact that such behavior involves speculation it will be subject to error.
- ▶ In the cases of error, the economy will have over-invested as the anticipated demand will not materialize.
- ▶ This will cause a recession and a process of liquidation.
- ▶ Both the boom and the bust are direct consequences of people's incentive to speculate on information related to future developments of the economy.

II.1. The New view

An example: the satellite industry

- ▶ 1990s: anticipation of high demand for bandwidth (related to the internet boom)
- ▶ The launching of telecommunication satellites exploded.
- ▶ In 1998, the satellite industry launched 150 satellites, 300 percent growth compared to 1993.
- ▶ The boom was not caused by a technological improvement nor an increase of current demand for bandwidth.
- ▶ The growth rate for demand for satellite bandwidth has been 31% between 1995 and 2003, while the supply of satellite bandwidth grew by 54% during this same timeframe
- ▶ At the beginning of the new millenium, demand failed to materialize *~> transponder glut.*
- ▶ The number of satellites launched went down to 75 in 2001 and 69 in 2003, against 150 in 1998.

II.1. The New view

An example: the satellite industry (*continued*)

- ▶ Variety, March 25, 1998 : *“**Sat glut in Arab skies:** With a glut of satellite transponder space available in the crowded Mideastern skies, the region looks set for a **cutthroat price war.** Arabic channels already have the multinational Arabsat as well as Eutelsat and Intelsat to choose from, and in late April these will be joined by Egypt’s Nilesat, which will have **72 transponders available for rental.**”*

II.1. The New view

An example: the satellite industry (*continued*)

- ▶ Satellite News, Nov 4, 2002 : *“Companies that purchased satellite transponder capacity during the **industry’s halcyon days** are now **struggling to resell the capability** in the face of falling market prices. One example is Globecom Systems [GCOM], a Hauppauge, N.Y.-based provider of end-to-end satellite-based communications services. To rid itself of carrying **\$400,000 in unused capacity each month**, Globecom reached a broad-based agreement with one of its vendors to cut the company’s future obligations for satellite bandwidth capacity.”*

II.1. The New view

News?

- ▶ Let's consider an exogenous variable, say productivity θ
- ▶ $Y = \theta F(K, H)$
- ▶ $\theta_t = \rho\theta_{t-1} + \varepsilon_t$
- ▶ $\theta_t = \rho\theta_{t-1} + \varepsilon_{t-q}$
- ▶ Signal $S_t = \varepsilon_t + \eta_t$, where η_t is a Gaussian white noise error term.
- ▶ **Note :** This is an example, News need not to be only about technology.

Roadmap of Part I

1. The News View
2. Informal and VAR Evidence
3. An Analytical Framework
4. Discussion / Extensions
5. Applications

II.2. Informal and VAR Evidence

- ▶ Here I display some informal evidence showing that there is evidence of news driven BC and liquidation cycles
- ▶ By liquidation cycle I mean the following sequence
 - × First phase of capital accumulation caused by rosy expectations.
 - × Second phase of downwards expectations revision.
- ▶ Then I show evidence for VARs

II.2. Informal and VAR Evidence

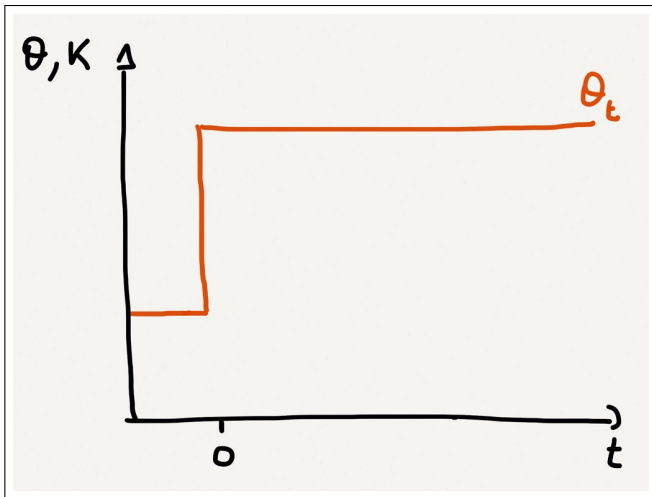
- Think of some stochastic G.E. model in which shocks to TFP are random walk.

$$\theta_t = \theta_{t-1} + \epsilon_t$$

- The state of the economy is summarized by $\frac{K}{\theta}$

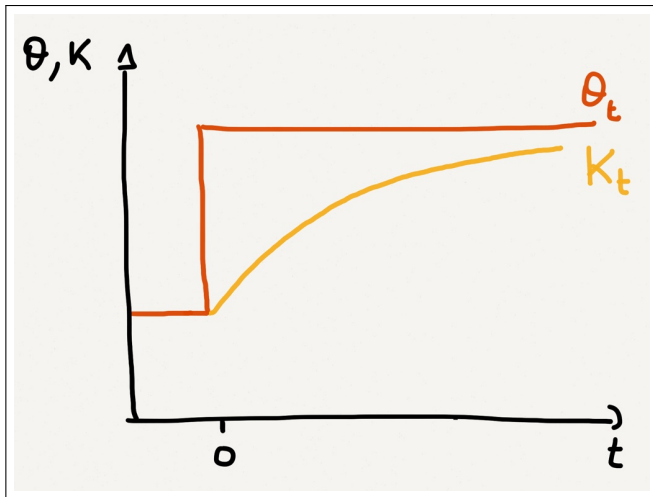
II.2. Informal and VAR Evidence

Figure 1: A surprising increase in θ



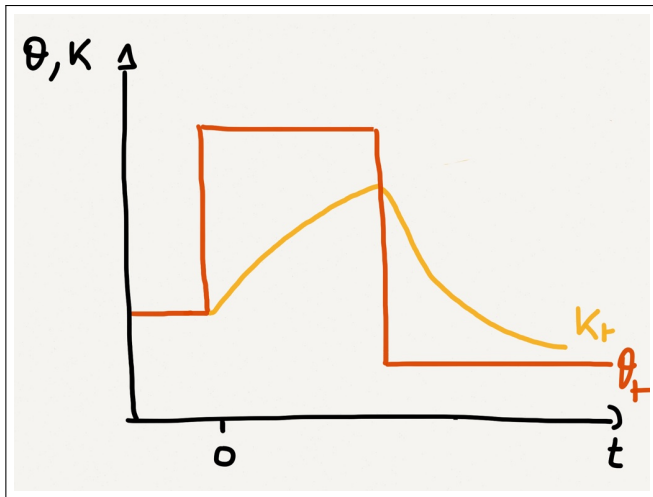
II.2. Informal and VAR Evidence

Figure 2: A surprising increase in θ



II.2. Informal and VAR Evidence

Figure 3: A surprising decrease in θ



II.2. Informal and VAR Evidence

Predictions for $\frac{K}{\theta}$

- ▶ If $\frac{K}{\theta}$ is low, the model predicts that employment should be high
- ▶ If $\frac{K}{\theta}$ is high, then employment should be low as there are low returns to capital accumulation + wealth effect.
- ▶ According to a “surprise” view, employment and $\frac{K}{\theta}$ should be strongly negatively correlated.
- ▶ That predictions holds in the same way if nominal rigidities are added (although it depends on the monetary rule)

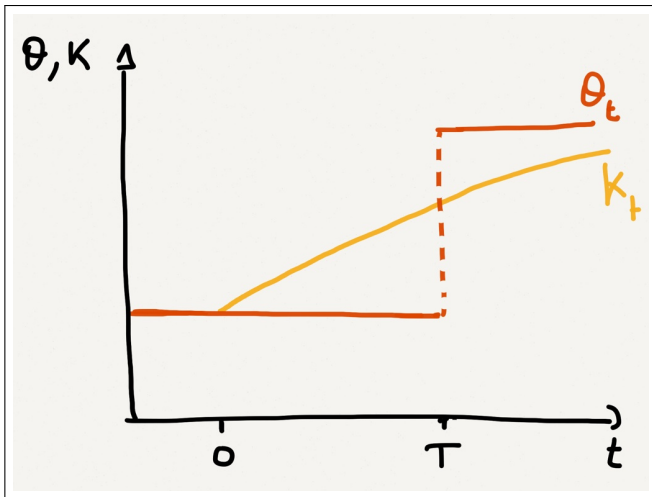
II.2. Informal and VAR Evidence

The News view

- ▶ The news view of business cycles suggest that employment and $\frac{K}{\theta}$ should have a modest positive correlation

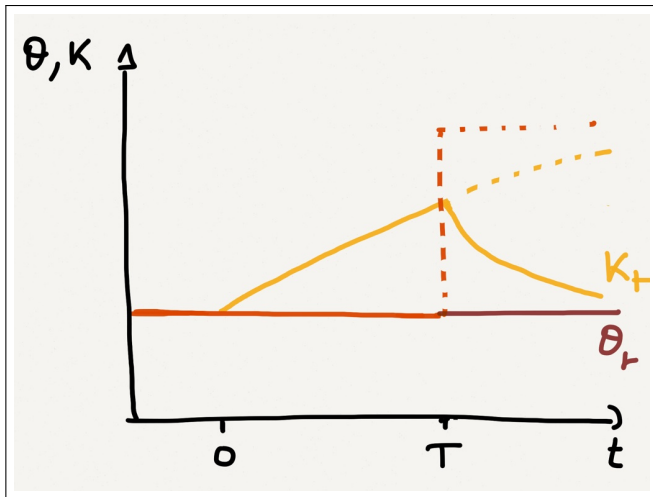
II.2. Informal and VAR Evidence

Figure 4: A news about future increase in θ



II.2. Informal and VAR Evidence

Figure 5: A downward revision of θ



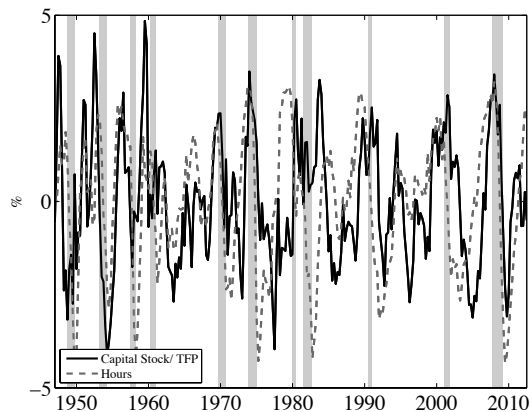
II.2. Informal and VAR Evidence

Predictions for $\frac{K}{\theta}$

- ▶ If employment booms arise because capital accumulation anticipates growth in TFP, then employment and $\frac{K}{\theta}$ should be positively correlated in booms.
- ▶ Major recessions in the news view of business cycles arise when $\frac{K}{\theta}$ is high and expectations no longer support such a high capital stock and this leads to a recession
- ▶ The second force should contribute to a negative correlation between employment and $\frac{K}{\theta}$
- ▶ Since on average agents should be right more often than wrong, it suggests that employment and $\frac{K}{\theta}$ should be positively correlated according to the news view.

II.2. Informal and VAR Evidence

Figure 6: $\frac{K}{\theta}$ and H , US 1948-2012



► correlation = .30

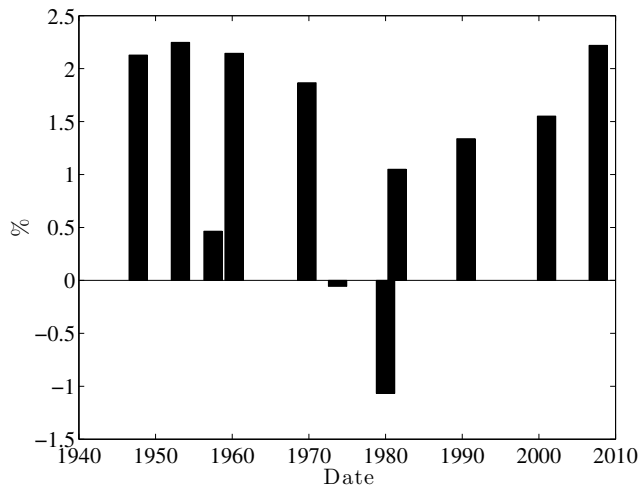
II.2. Informal and VAR Evidence

Recession episodes

- ▶ In the “surprise” view, recessions are caused by negative technological shock.
- ▶ Major recessions are therefore caused by major shocks to TFP, that are not correlated with the level of $\frac{K}{\theta}$.
- ▶ In the news view, major recessions occur when $\frac{K}{\theta}$ is high and expectations revised.
- ▶ One should expect $\frac{K}{\theta}$ to be *high* the periods before a major recession.

II.2. Informal and VAR Evidence

Figure 7: Mean HP deviation of $\frac{K}{\theta}$ the year of a NBER peak



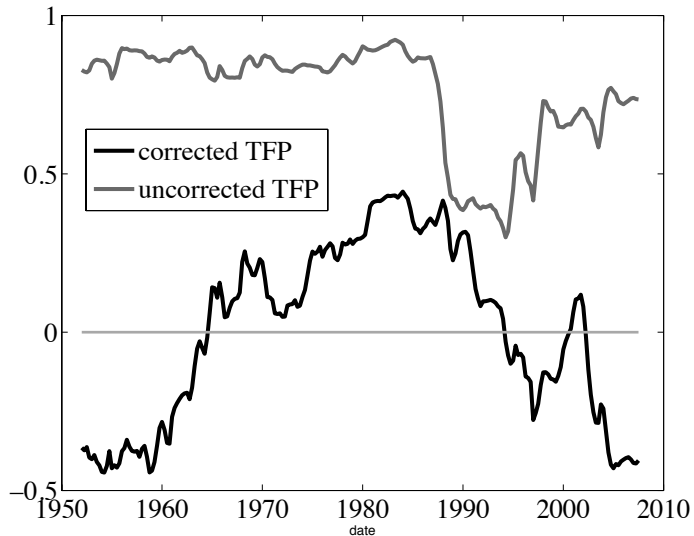
II.2. Informal and VAR Evidence

Shocks

- ▶ Why are News/Revision shocks not only possible but also of interest?
- ▶ Because technological surprises are not the story (and no obvious observed “demand shocks”)

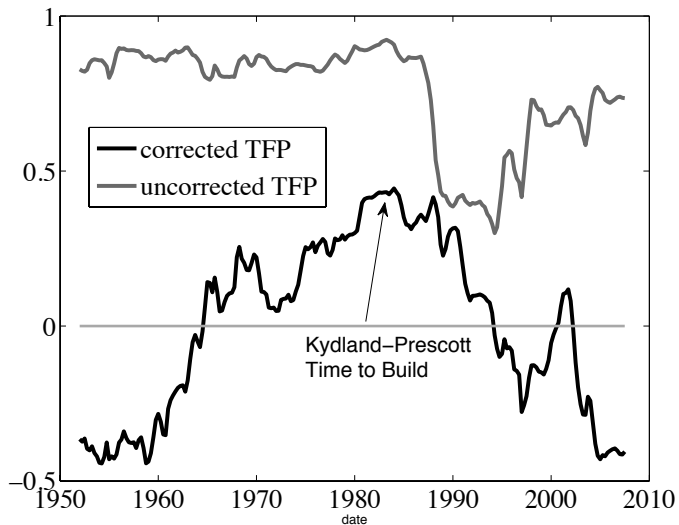
II.2. Informal and VAR Evidence

Typical TFP shocks are not good candidates for the post Volcker period



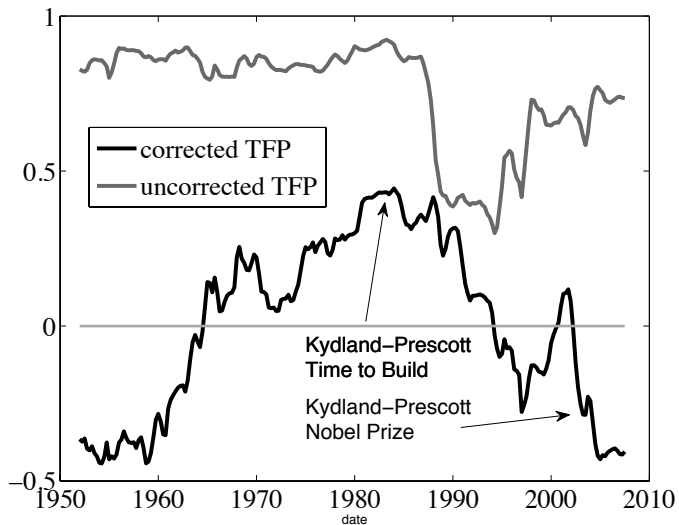
II.2. Informal and VAR Evidence

Typical TFP shocks are not good candidates for the post Volcker period



II.2. Informal and VAR Evidence

Typical TFP shocks are not good candidates for the post Volcker period



II.2. Informal and VAR Evidence

Neither are IST shocks

- ▶ If IST shocks, investment is cheap in booms, expensive in recessions.
- ▶ Various measures of the relative price of investment, deflating with core CPI, correlations with Hours:

Variable	1960Q1-2012Q3	Post-Volcker
Fixed I	0.42	0.76
Struct.I	0.44	0.75
Equip.I	-0.25	0.17
Resid.I	0.70	0.80
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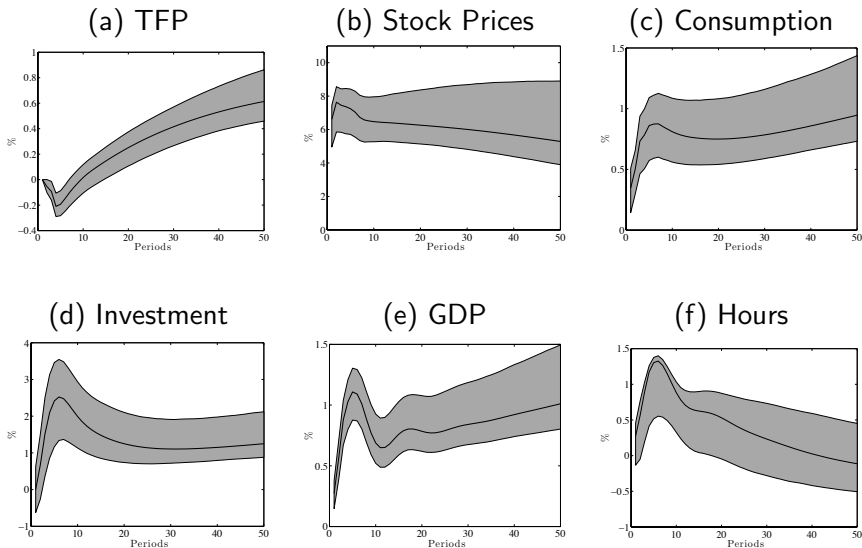
II.2. Informal and VAR Evidence

Identification of a News Shock in VARs

- ▶ View the data as being generated by many shocks among which:
 - × Only 2 have a permanent effect on TFP (surprise and news)
 - × Only 1 has a impact effect on TFP (surprise)
- ▶ A news shock is therefore a shock
 - × That has a permanent effect of TFP
 - × That has no impact effect on TFP
- ▶ What is the response of the economy to such a shock (US data)

II.2. Informal and VAR Evidence

Response to a TFP news



Roadmap of Part I

1. The News View
2. Informal and VAR Evidence
3. [An Analytical Framework](#)
4. Discussion / Extensions
5. Applications

II.3. An Analytical Framework

Overview

- ▶ A quite simple analytical model
- ▶ I show that investment is explicitly driven by future profit opportunities
- ▶ I show how boom/bust cycles can be generated
- ▶ The discipline for some assumptions and the choice of functional forms: tractability, not quantitative relevance.

II.3. An Analytical Framework

Households



$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\ln(C_t) + \nu \cdot (\bar{L} - L_t^T) \right]$$

► The household can:

- × buy some or all of its consumption on the market
- × supply labor time to the market
- × produce household consumption
- × buy one-period bonds
- × trade in firm shares.



$$C_t^M + B_{t+1} + \sum_{i=1}^N \mathcal{P}_{it} Z_{it+1} = w_t L_t^M + B_t(1 + r_t) + \sum_{i=1}^N \mathcal{P}_{it} Z_{it} + \sum_{i=1}^N d_{it} Z_{it+1}$$

► Home production: $C_t^H = \alpha L_t^H$

► $C_t = C_t^M + C_t^H$ and $L_t^T = L_t + L_t^H$

II.3. An Analytical Framework

Production side

- ▶ Set of intermediate sectors $i = 1, \dots, N$
- ▶ final good sector that aggregates the intermediate goods into a market consumption good according to:

$$C_t^M = \left[\sum_{i=1}^N \frac{1}{N} (\theta_{it} X_{it})^\phi \right]^{\frac{1}{\phi}}, \quad \phi < 1$$

- ▶ Intermediate sectors and final market consumption good sector are competitive.
- ▶ Market consumption good is the numéraire
- ▶ Price of the intermediate goods: P_{it}

II.3. An Analytical Framework

Intermediate goods (1)

- ▶ The representative firm in sector i produces the intermediate good X_{it} :

$$X_{it} = K_{it}^{\gamma} (L_{it}^P)^{1-\gamma}, \quad 0 < \gamma \leq 1$$

- ▶ The firm also hires labor L_{it}^I to build up its capital stock:

$$K_{it+1} = I_{it} + (1 - \delta)K_{it},$$

with $I_{it} = \ln(L_{it}^I)$

- ▶ Let dividends be $d_{it} = P_{it}X_{it} - w_t L_{it}$

II.3. An Analytical Framework

Intermediate goods (2)

- The firm maximizes expected discounted stream of dividends:

$$\max_{L_t^P, L_t^I} E_0 \sum_{t=0}^{\infty} \frac{1}{1 + R_{0,t}} (P_{it} X_{it} - w_t L_{it})$$

subject to

$$X_{it} = K_{it}^{\gamma} (L_{it}^P)^{1-\gamma}, \quad 0 < \gamma \leq 1$$

$$K_{it+1} = \ln(L_{it}^I) + (1 - \delta)K_{it}$$

- Firms are assumed to finance themselves by retained earnings
- Shares in each sector are normalized to 1.

II.3. An Analytical Framework

Walrasian equilibrium

Result 1

The equilibrium is characterized by a constant wage rate $w_t = \alpha$ and a constant interest rate $(1 + r_t) = \frac{1}{\beta}$.

II.3. An Analytical Framework

Firm Behavior (1)

- ▶ This result makes the problem of the intermediate firm easier to solve.
- ▶ Optimal investment:

$$I_{it} = \ln \left(\frac{\beta}{\alpha} \gamma (1 - \gamma)^{\frac{1-\gamma}{\gamma}} E_t \sum_{j=0}^{\infty} (\beta(1 - \delta))^j P_{it+j+1}^{\frac{1}{\gamma}} \right)$$

- ▶ Investment is purely forward looking
- ▶ Need to build capacity in anticipation of the economy's future needs.

II.3. An Analytical Framework

Firm Behavior (2)

- Optimal labor demand:

$$\begin{aligned}L_{it} &= L_{it}^P + L_{it}^I \\&= K_{it} \left(\frac{(1-\gamma)P_{it}}{\alpha} \right)^{\frac{1}{\gamma}} \\&\quad + \frac{\beta}{\alpha} \gamma (1-\gamma)^{\frac{1-\gamma}{\gamma}} E_t \sum_{j=0}^{\infty} (\beta(1-\delta))^j P_{it+j+1}^{\frac{1}{\gamma}}\end{aligned}$$

- Employment itself is partially determined by expectations of future capital prices, that signal future (expected) needs of capital.

II.3. An Analytical Framework

Firm Behavior (3)

- ▶ Intermediate firms need to form expectations about the future price of their output P_{it+j} :

$$P_{it+j} = \frac{1}{N} \left(\frac{C_{t+j}^M}{X_{it+j}} \right)^{1-\phi} \theta_{it+j}^\phi$$

- ▶ To form expectations of future prices, firms will want to have information relevant for predicting aggregate consumption, aggregate production in their sector and technological change in their sector.
- ▶ Some of this information may be contained in current news (for example signals about future changes in the θ 's).
- ▶ This is in general a hard problem to solve (see later), but tractable in the case $\phi = 0$.

$$C_t^M = \left[\sum_{i=1}^N \frac{1}{N} (\theta_{it} X_{it})^\phi \right]^{\frac{1}{\phi}}$$

II.3. An Analytical Framework

The case $\phi = 0$

$$C_t^M = \left[\sum_{i=1}^N \frac{1}{N} (\theta_{it} X_{it})^\phi \right]^{\frac{1}{\phi}}$$

- ▶ When $\phi = 0$, P_{kt+j} writes

$$P_{kt+j} = \frac{1}{N} \frac{C_{t+j}^M}{X_{kt+j}} = \frac{1}{N} \frac{\left(\prod_{i=1}^N (X_{it+j})^{\frac{1}{N}} \right)}{X_{kt+j}} \left(\prod_{i=1}^N (\theta_{it+j})^{\frac{1}{N}} \right)$$

- ▶ The firm needs predict future market consumption and the future output level that will be offered in his sector (its market share, the demand for its own good).
- ▶ However..., at the symmetrical equilibrium

$$P_{kt+j} = \Theta_{t+j}$$

II.3. An Analytical Framework

News and noise

- ▶ Technology index Θ follows an autoregressive process:

$$\Theta_t = \bar{\Theta} + \rho\Theta_{t-1} + \epsilon_{t-q}, \quad 0 \leq \rho < 1$$

- ▶ ϵ_t is a Gaussian white noise process.
- ▶ Signal

$$S_t = \epsilon_t + \eta_t$$

- ▶ η_t is a Gaussian white noise error term.
- ▶ Let's study the response of the economy to news and noise in that tractable case where $\phi = 0$ and $\gamma = 1$, so that L is fully forward looking.

$$X_{it} = K_{it}^{\gamma} (L_{it}^P)^{1-\gamma}$$

II.3. An Analytical Framework

When $\gamma = 1$ (1)

- Labour demand writes:

$$L_{it} = \frac{\beta}{\alpha} E_t \sum_{j=0}^{\infty} (\beta(1 - \delta))^j P_{it+j+1}$$

- Let $\psi = \frac{\sigma_{\epsilon}^2}{\sigma_{\eta}^2 + \sigma_{\epsilon}^2}$ captures the information content of the signal S_t with respect to ϵ_t

II.3. An Analytical Framework

When $\gamma = 1$ (2)

- Relative deviation of L from SS:

$$\begin{aligned}\hat{L}_t &= \frac{(1 - (1 - \delta)\beta)}{(1 - \rho(1 - \delta)\beta)} \times \\ &\quad \left[\sum_{j=0}^{q-1} ((1 - \delta)\beta)^{q-j-1} \psi \epsilon_{t-j} + \sum_{j=q}^{\infty} \rho^{j-q+1} \epsilon_{t-j} \right] \\ &\quad + \left[\sum_{j=0}^{q-1} ((1 - \delta)\beta)^{q-j-1} \psi \eta_{t-j} \right]\end{aligned}$$

- In comparison

$$\hat{\Theta}_t = \sum_{j=0}^{\infty} \rho^j \epsilon_{t-q-j}$$

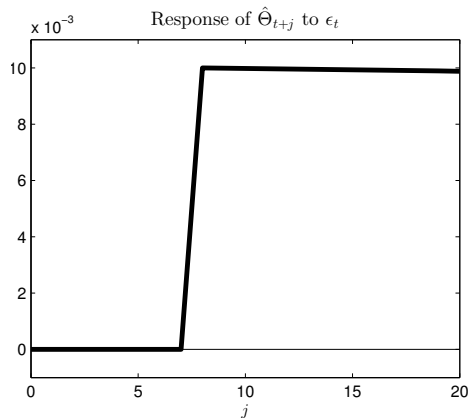
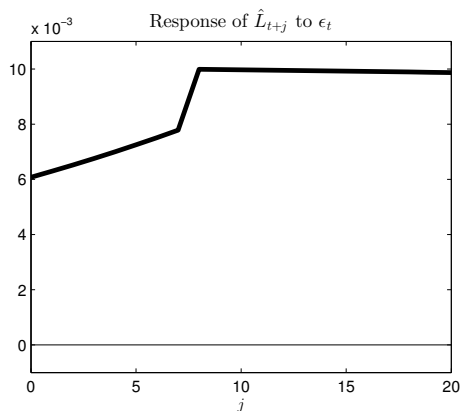
II.3. An Analytical Framework

IRF (1)

- ▶ Numerical illustration: $\beta = .99$, $\delta = .025$, $\rho = .999$, $q = 8$, $\psi = .8$, $\alpha = 25$ and $\kappa = 0.1$.
- ▶ Consider successively news and noise shocks.

II.3. An Analytical Framework

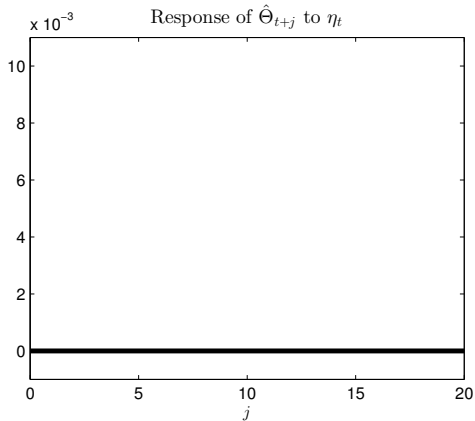
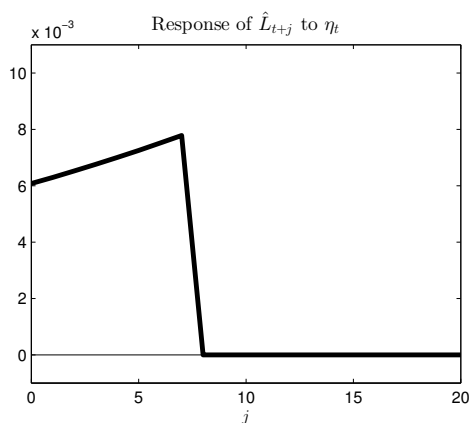
IRF to news



- On receiving the news employment starts to increase immediately, while $\hat{\Theta}$ does not change

II.3. An Analytical Framework

IRF to noise



- A key feature of this model is that it creates potentially large reversals in employment as the result of agents re-evaluating their information.

II.3. An Analytical Framework

Asset prices

- Define

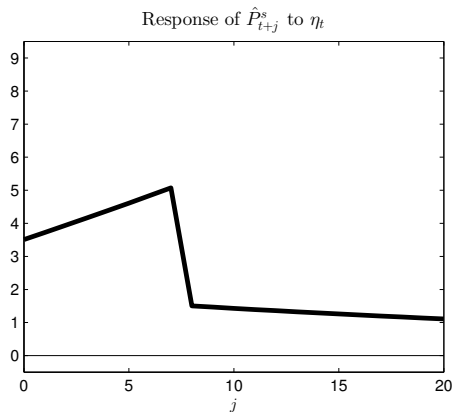
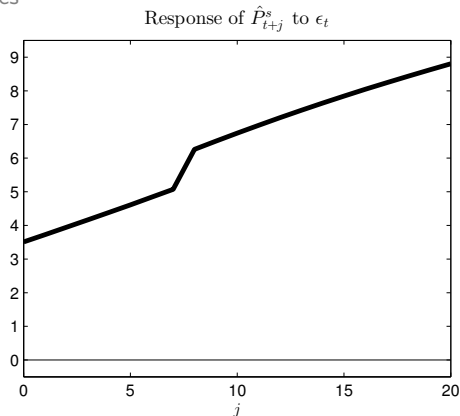
$$\mathcal{P}_{it} = E_t \sum_{j=0}^{\infty} \beta^j d_{it+j} = E_t \sum_{j=0}^{\infty} \beta^j (P_{it+j} K_{it+j} - w_t L_{it+j})$$

- First-order approximation:

$$\begin{aligned} \hat{\mathcal{P}}_t &= \sum_{k=0}^{q-1} \left\{ \tilde{K} \frac{\beta^{q-k}}{1-\rho\beta} \psi + [1 + \alpha(1-\delta)\tilde{L}] B_{k-1} \psi \right\} \epsilon_{t-k} \\ &+ \sum_{k=q}^{\infty} \left\{ \tilde{K} \frac{\rho^{k-q}}{1-\rho\beta} + [1 + \alpha(1-\delta)\tilde{L}] [A_{k-1} + B_{k-1}(1-\delta)^{k-q} \psi] \right\} \epsilon_{t-k} \\ &+ \sum_{k=0}^{q-1} \left\{ \tilde{K} \frac{\beta^{q-k}}{1-\rho\beta} \psi + [1 + \alpha(1-\delta)\tilde{L}] B_{k-1} \psi \right\} \eta_{t-k} \\ &+ \sum_{k=q}^{\infty} \left\{ [1 + \alpha(1-\delta)\tilde{L}] B_{q-1} (1-\delta)^{k-q} \psi \right\} \eta_{t-k} \end{aligned}$$

II.3. An Analytical Framework

Stock prices



- **Remark:** If L bears adjustment costs, the stock price would be the best indicator of the agents' information and would be the first variable to move in response to news. (key insight for identification of news)

Roadmap of Part I

1. The News View
2. Informal and VAR Evidence
3. An Analytical Framework
4. Discussion / Extensions
5. Applications

II.4. Discussion / Extensions

The non-trivial information processing problem, and the interpretation of errors (1)

$$C_t^M = \left[\sum_{i=1}^N \frac{1}{N} (\theta_{it} X_{it})^\phi \right]^{\frac{1}{\phi}}$$

- ▶ With $\phi = 0$, the model boils down to a boring expectation of an exogenous variable Θ .
- ▶ When $\phi \neq 0$, the prediction problem faced by firms becomes much more involved.
- ▶ To see this, let's consider another specific version of the model:
 - × $\phi \neq 0$
 - × $\delta = 1$
 - × $\theta_{jt} = \bar{\theta} + \epsilon_{jt-1}$ (*iid*)
 - × $S_{it} = \epsilon_{jt} + \eta_{it}$ (*one period news*)

II.4. Discussion / Extensions

The non-trivial information processing problem, and the interpretation of errors (2)

- We obtain

$$E_{jt} P_{jt+1} = E_{jt} \left[\frac{\sum_{i=1}^N (\epsilon_{it+1} \ln (E_{it} [\frac{\beta}{\alpha} P_{it+1}]))^{\phi}}{X_{jt}} \right]^{1-\sigma} \epsilon_{jt+1}^{\phi}$$

- Non linear problem
- Complementarity structure: firms in one sector will want to increase their production if they expect others to increase their production, regardless of the actual news received.
- There is a rational expectation solution with common information
- Errors by some firms lead to errors by other firms: if limited ability to process information, cycles can occur.
- News shocks generally coined as “augmented TFP” shocks, but this is a restricted view

II.4. Discussion / Extensions

Recessions as Liquidation Periods (1)

- ▶ Booms are periods in which expectations might be rosy (because of the noise).
- ▶ The post-bust periods (after expectation revisions) are periods of *liquidation*, as investment falls and capital is depleted.
- ▶ However, I and L simply return to SS because we have a simplified setup.
- ▶ Model modification:
 - × $C_t^M = \Theta_t X_t - \kappa \frac{X_t^2}{2}$ (DRS)
 - × Keep noisy signals on Θ
 - × $I_{it} = 2(L_{it}^I)^{\frac{1}{2}}$

II.4. Discussion / Extensions

Recessions as Liquidation Periods (2)

- Solution is

$$K_{t+1} = \lambda K_t + \frac{\beta \lambda}{1 - \delta} \sum_{j=0}^{\infty} (\beta \lambda)^j E_t \Theta_{t+1+j}$$

- and

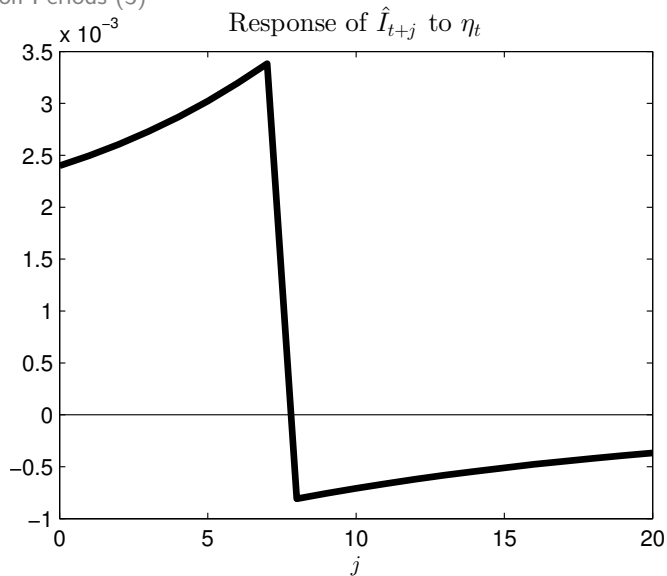
$$\hat{l}_t = \Omega \sum_{j=0}^{\infty} \phi_j \eta_{t-j}$$

where

$$\Omega \equiv \frac{\beta (\beta \lambda)^{q+1}}{(1 - \delta) (1 - \rho \beta \lambda) (1 - \beta \lambda^2)}$$
$$\phi_j \equiv \begin{cases} \lambda^j \left\{ \lambda \left[(\beta \lambda^2)^{-1-j} - 1 \right] - (1 - \delta) \left[(\beta \lambda^2)^{-j} - 1 \right] \right\} \psi & \text{if } j < q \\ -(1 - \delta - \lambda) \left[(\beta \lambda^2)^{-q} - 1 \right] \lambda^j \psi & \text{if } j \geq q \end{cases}$$

II.4. Discussion / Extensions

Recessions as Liquidation Periods (3)



II.4. Discussion / Extensions

The efficiency of news-driven booms and busts

- ▶ In that basic model, fluctuations are constrained-efficient.
- ▶ But our core mechanism does to rely on constrained efficiency.
- ▶ Assume a constant labor market wedge: $w = \alpha(1 + \tau)$, $\tau > 0$
- ▶ Same positive implications
- ▶ Different normative ones: in a noise driven boom, more employment and consumption $\rightsquigarrow$ the economy gets closer to an efficient outcome.
- ▶ The boom is a good period even if expectations are not met eventually.
- ▶ (This is *not* an unconditional statement)
- ▶ But one can think of a wedge that varies with the business cycle.

II.4. Discussion / Extensions

Expectations of new markets as a form technological news

- ▶ News need not to be about future productivity.
- ▶ Assume

$$C_t = N_t^\xi \left(\sum_{i=1}^{N_t} X_{it}^\sigma \right)^{\frac{1}{\sigma}}, \quad \sigma < 1$$

- ▶ N_t is now the driving force, and expands over time, with news and noise.
- ▶ Symmetric equilibrium:

$$C_t = N_t^{\xi + \frac{1}{\sigma}} \bar{X}_t$$

- ▶ Hence, much of the literature on technological news can be re-interpreted as models with news of expanding market.
- ▶ Assume $\xi = -\frac{1}{\sigma}$, monopoly power for each intermediate and a setup cost to implement a new variety: totally inefficient business cycles driven by changes in N_t .

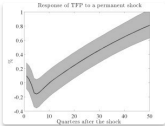
II.3. To Go Further

fportier.wordpress.com/current-work/news-driven-business-cycles-insights-and-challenges-data-and-codes/ — News Driven Business Cycles: Insights and Challenges - Data and Codes | (macro... Lecteur

(macro)Economics live from Toulouse
Franck Portier's professional page

HOME BRIEF CURRICULUM VITAE CURRENT WORK DISCUSSIONS PUBLISHED RESEARCH TEACHING/ENSEIGNEMENT

News Driven Business Cycles: Insights and Challenges – Data and Codes



Here are gathered data and codes for the replication of all the estimations and simulations of the paper "News Driven Business Cycles: Insights and Challenges" co-authored with Paul Beaudry and that is forthcoming in the Journal of Economic Literature.

Data are in Excel files and codes are Matlab. Codes are commented so that they should be understandable. None of the codes have been written with the aim of being elegant nor efficient. Therefore, they are not.

Those codes are distributed in the hope that they will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

The whole archive file "BP JEL Public codes and data" with data and code can be obtained [here](#).

- **JELbase.mat** is the matlab data file that gathers all the data we need. It is generated from the file **MakeJELBase.m** that is located in the directory **data**. In that directory are gathered all the Excel files needed.
- The directory **Section1model** contains the code that computes the IRF of the theoretical model.
- The directory **Var** contains all the codes to estimate and the Monte Carlo simulations, for the models with 2, 3 and 4 variables. **Factors** is a subdirectory that estimates a model a VECM with 7 variables that includes the two first factors of Forni, Gambetti and Sala (2014, Economic Journal). Those factors, kindly provided by FGS, are in an Excel file.
- The directory **ReducedFormEvidence** contains the code that computes the statistics used in the "reduced form evidence" of the paper.

Roadmap of Part I

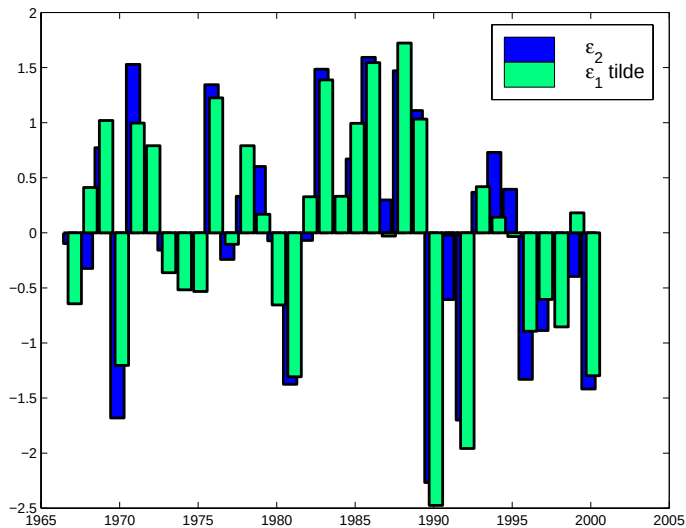
1. The News View
2. Informal and VAR Evidence
3. An Analytical Framework
4. Discussion / Extensions
5. Applications

II.5 Applications: Japan Lost Decade

- ▶ How to explain the “bubble burst” of the early 90’s, then the slow growth of the 90’s
- ▶ A downward revision of TFP growth (that is left to be explained)
- ▶ This is what the data suggest.

II.5 Applications: Japan Lost Decade

Estimated News Shocks



II.5 Applications: Japan Lost Decade

- ▶ What do we get from this figure?
- ▶ Two stock market shocks at the beginning of the 1990s, that where possibly the consequence of bad news about future TFP , explain most of TFP changes in the 1990s and about half of the stock market variations.
- ▶ Gives some rationale to Hayashi and Prescott (2002) intuition:

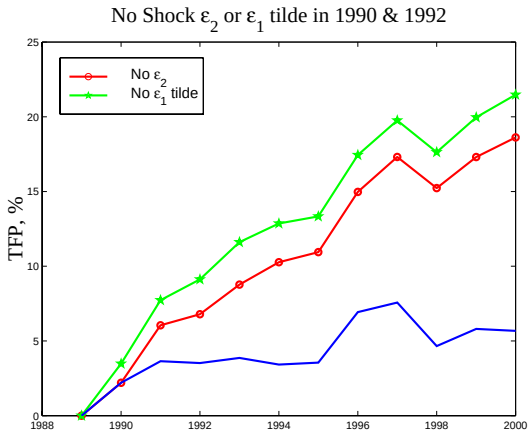
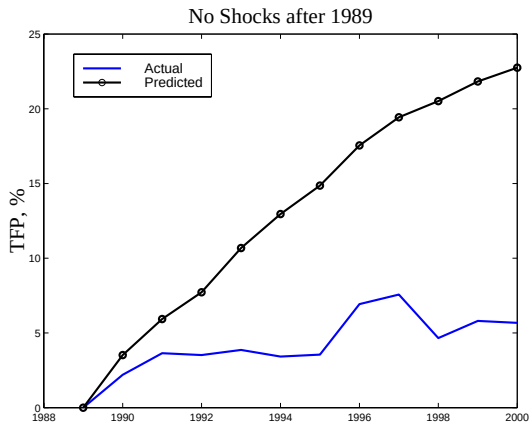
II.5 Applications: Japan Lost Decade

“In examining the virtual stagnation that Japan began experiencing in the early 1990s, we find that the problem is not a breakdown of the financial system [...]. The problem is low productivity growth. [...] We said very little about the “bubble” period of the late 1980s and early 1990s, a boom period when property prices soared, investment as a fraction of GDP was unusually high, and output grew faster than in any other years in the 1980s and 1990s. We think the unusual pickup in economic activities, particularly investment, was due to an anticipation of higher productivity growth that never materialized. To account for the bubble period along these lines, we need to have a model where productivity is stochastic and where agents receive an indicator of future productivity.” (italics added by myself)

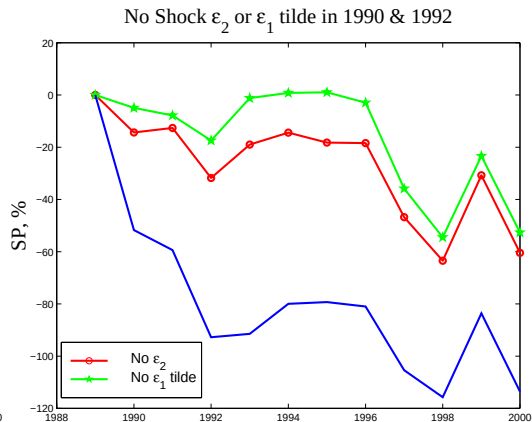
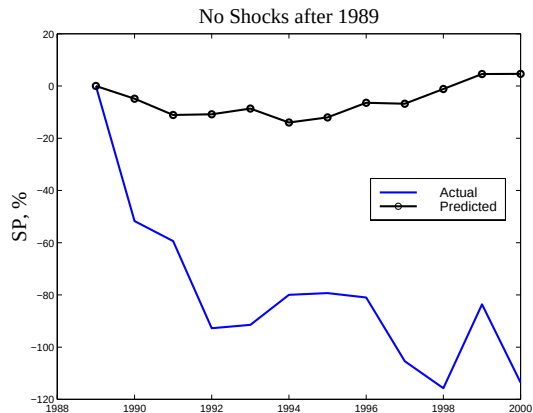
(page 227-228), Hayashi & Prescott, Japan's Lost Decade, RED, 2002

II.5 Applications: Japan Lost Decade

Historical Decomposition of the 1990s



Historical Decomposition of the 1990s



II.5 Applications: A theory of international comovements

International Business Cycle Comovements

- ▶ At business cycle frequencies, Y , C , I , H are:
 - positively correlated within countries,
 - pairwise positively correlated among countries.
- ▶ Local technology shocks imply reallocation of mobile inputs
~> negative comovements unless almost perfectly correlated shocks.
- ▶ As technology shocks are not 'global enough', 'Demand' shocks and frictions are needed for synchronization business cycles.

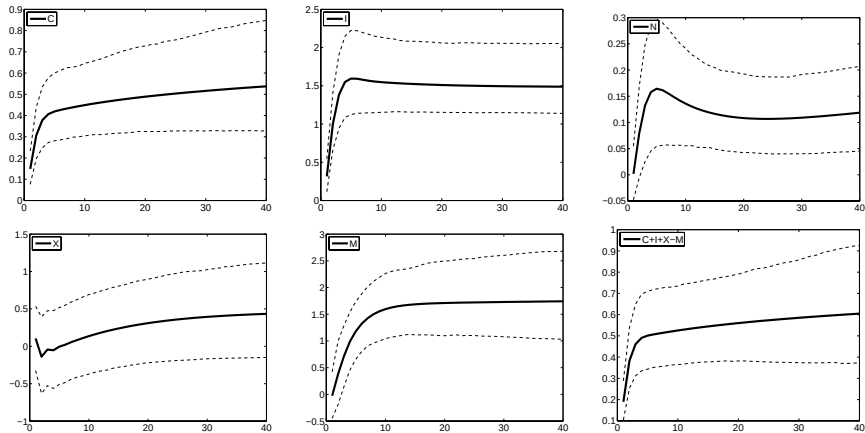
II.5 Applications: A theory of international comovements

- ▶ We show theoretically that shocks to expectations (news) are indeed creating international comovements. (*even if the realization is local, information is global*)
- ▶ We identify in the data a particular news (TFP news), and show that it is internationally transmitted (US/Canada, Germany/Austria).
- ▶ We propose a model that is able to replicate the qualitative features of the data.

II.5 Applications: A theory of international comovements

A news shock triggers an expansion in the US...

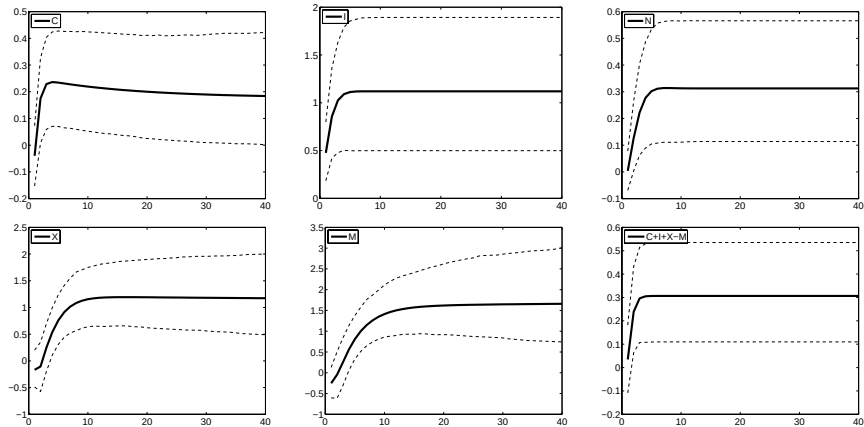
Response to a news shock, USA



II.5 Applications: A theory of international comovements

...as well as in Canada.

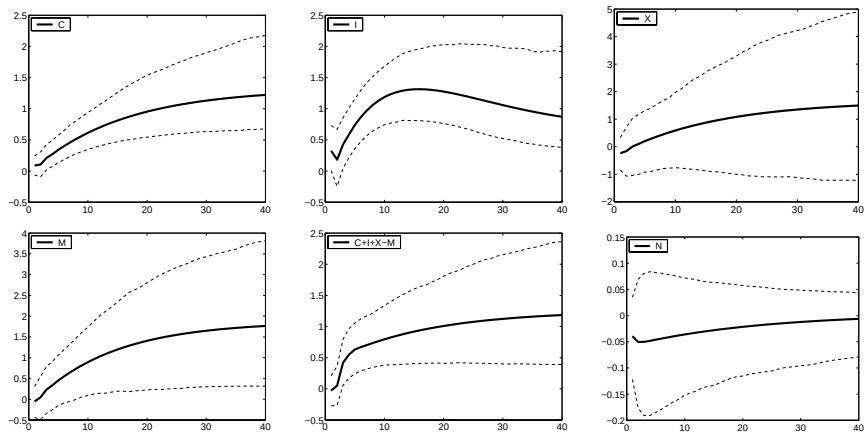
Response of Canadian aggregates to a news on US TFP



II.5 Applications: A theory of international comovements

Again a news shock triggers an expansion in Germany...

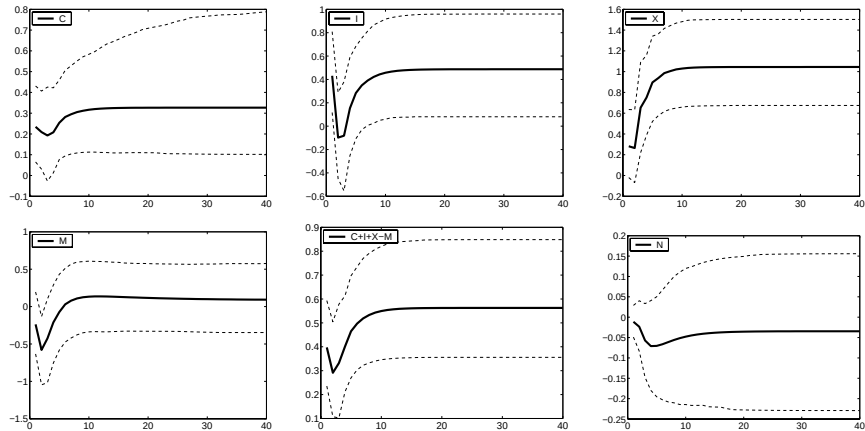
Response to a news shock, Germany



II.5 Applications: A theory of international comovements

... as well as in Austria.

Response of Austrian aggregates to a German News Shock



Part III : Liquidations

III.1. Motivations

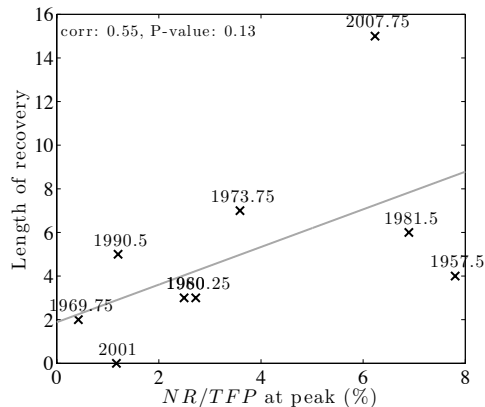
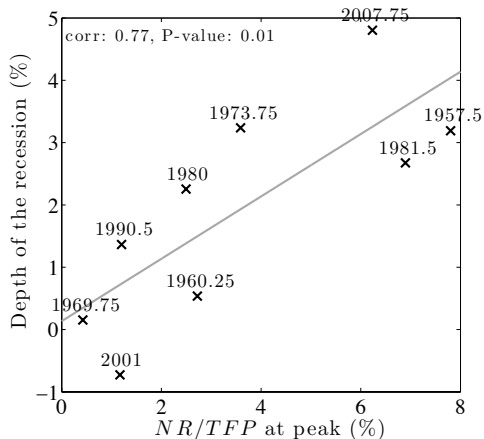
Recessions

- ▶ Recessions often happen after periods of rapid accumulation of houses, consumer durables and business capital.
- ▶ Obvious examples: The Great Depression, The dot com crash, the Asian crisis of 1997, etc...
- ▶ Systematic look at the US business cycle:

III.1. Motivations

Recessions

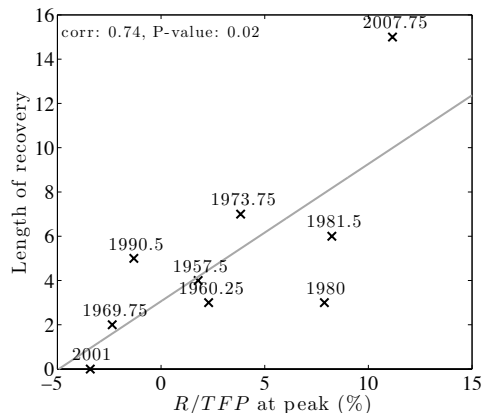
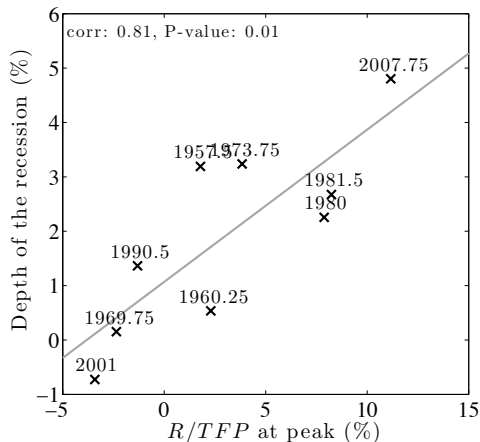
Figure 8: Depth of recess. and length of recov. vs. cumulated invest., Non Residential Investment



III.1. Motivations

Recessions

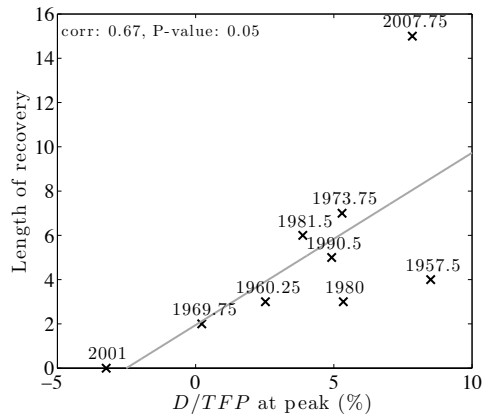
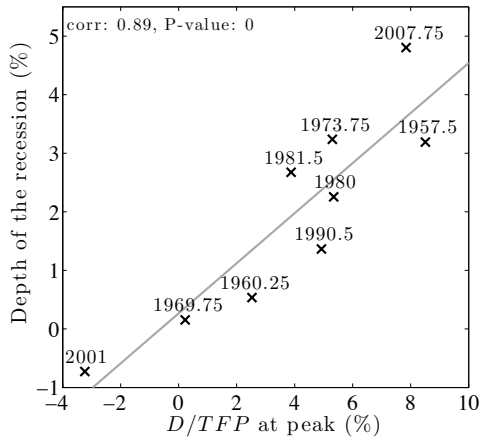
Figure 9: Depth of recess. and length of recov. vs. cumulated invest., Residential Investment



III.1. Motivations

Recessions

Figure 10: Depth of recess. and length of recov. vs. cumulated invest., Durable Goods



III.1. Motivations

Recessions

- ▶ Two opposite views of economic policy in recessions
 - × HAYEK
 - × KEYNES

III.1. Motivations

The Liquidationist View (FRIEDRICH HAYEK)

- ▶ Recessions are needed to cleanse the economy.
- ▶ Gvt spendings, aggregate demand management only delays necessary adjustment
- ▶ J. SCHUMPETER, The Economics of the Recovery Program, 1934 :
 - × *“Any revival which is merely due to artificial stimulus leaves part of the work of depressions undone”,*

III.1. Motivations

The Aggregate Demand View (JOHN MAYNARD KEYNES)

- ▶ Recessions are periods of insufficient demand
- ▶ Activist fiscal policy is needed
- ▶ J.M. KEYNES, The General Theory of Employment, Interest and Money, 1936:
 - × *“If the Treasury were to fill old bottles with bank-notes*
 - × *bury them at suitable depths in disused coal-mines which are then filled up to the surface with town rubbish,*
 - × *and leave it to private enterprise on well-tryed principles of laissez-faire to dig the notes up again [...],*
 - × *there need be no more unemployment*
 - × *and, with the help of repercussions, the real income of the community, and its capital wealth, would probably become a good deal greater than it actually is”*

III.2. Another Perspective

- ▶ The two views are not mutually exclusive
- ▶ Liquidations can produce periods where the economy functions particularly inefficiently.
- ▶ Many socially desirable trades between individuals may remain unexploited.
- ▶ In this sense, liquidations can cause recessions characterized by deficient aggregate demand.
- ▶ Some stimulative policies may remain desirable even if they postpone a recovery.

III.2. Another Perspective

Main Ingredients

- ▶ Environment with decentralized markets, flexible prices and search frictions.
- ▶ Two imperfections:
 - × Labor market matching friction in the spirit of Diamond-Mortensen-Pissarides,
 - × Adverse selection in the insurance market : unemployment risk is not insurable.

III.2. Another Perspective

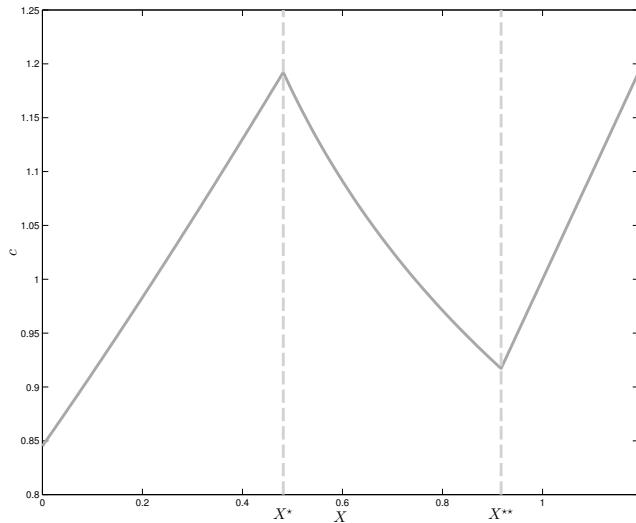
Main Mechanism

- ▶ Environment with decentralized markets, flexible prices and search frictions.
- ▶ If the economy finds itself with an excess of accumulated goods (durables and/or capital goods):
 - × Consumers and firms will spend less because they already have a lot,
 - × Firms will hire less as demand is low
 - × Consumers will consume less by fear of being unemployed,
 - × Spendings will therefore be low
 - × etc...
- ▶ There will be socially excessive precautionary savings
- ▶ Government spending can boost mutually beneficial trades ...
- ▶ ... but it will postpone the recovery by slowing down the liquidation process.

III.2. Another Perspective

Main Result

Figure 11: Consumption as function of X .

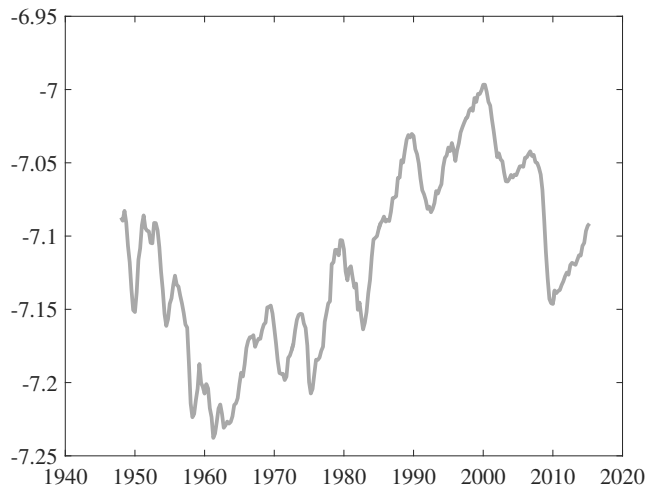


Part IV : Economic Cycle: Some Further Evidence

IV. Further Evidence

Common Way of Looking at the Data

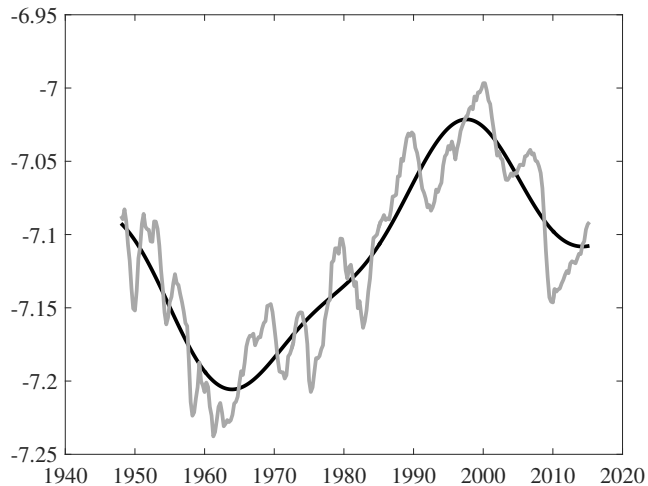
Figure 12: Total Hours



IV. Further Evidence

Common Way of Looking at the Data

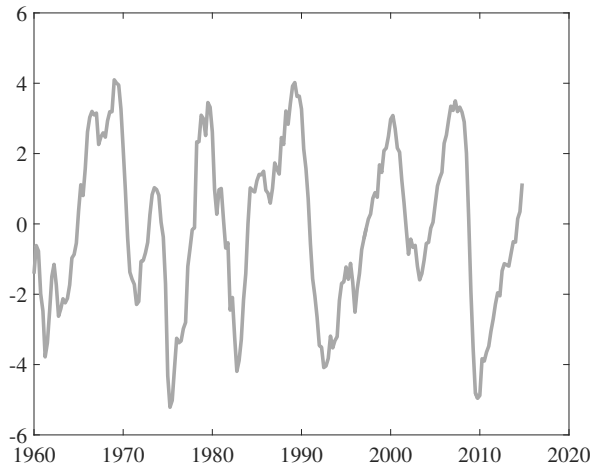
Figure 13: Total Hours and Trend (High-pass filter, 80 quarters)



IV. Further Evidence

Common Way of Looking at the Data

Figure 14: Cyclical Component of Total Hours



IV. Further Evidence

Common Way of Looking at the Data

- Fit an AR(2) (standard AIC-BIC criteria)

- Ordinary Least-squares Estimates

Dependent Variable = x

R-squared = 0.9362

Durbin-Watson = 2.2217

Nobs, Nvars = 220, 3

Variable	Coefficient	t-statistic	t-probability
constant	-0.000712	-0.017705	0.985891
x(-1)	1.418969	23.746422	0.000000
x(-2)	-0.481165	-8.054144	0.000000

Max Eigenvalue Modulus

AR :0.85849 [0.85849]

Autocor : .95

IV. Further Evidence

Common Way of Looking at the Data

- ▶ $X_t = AX_{t-1} + \varepsilon_t$
- ▶ $\lambda_1, \dots, \lambda_n$ eigenvalues of A
- ▶ Asymptotic behavior of X given by the largest (in modulus) eigenvalue $\lambda_{\max}$
- ▶ $|\lambda_{\max}| < 1$: stable
- ▶ $|\lambda_{\max}| > 1$: explosive
- ▶ λ complex : oscillations

IV. Further Evidence

Common Way of Looking at the Data

- ▶ $x_t = \alpha_0 + \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \varepsilon_t$
- ▶ Dynamic simulation :

$$\hat{x}_1 = x_1$$

$$\hat{x}_2 = x_2$$

$$\hat{x}_t = \hat{\alpha}_0 + \hat{\alpha}_1 x_{t-1} + \hat{\alpha}_2 x_{t-2}$$

- ▶ Simulation as of date T_0 :

$$x_1^S = x_1$$

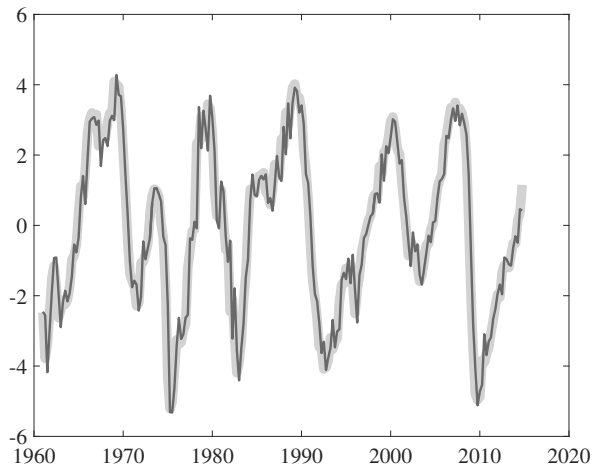
$$x_2^S = x_2$$

$$x_t^S = \hat{\alpha}_0 + \hat{\alpha}_1 x_{t-1}^S + \hat{\alpha}_2 x_{t-2}^S$$

IV. Further Evidence

Common Way of Looking at the Data

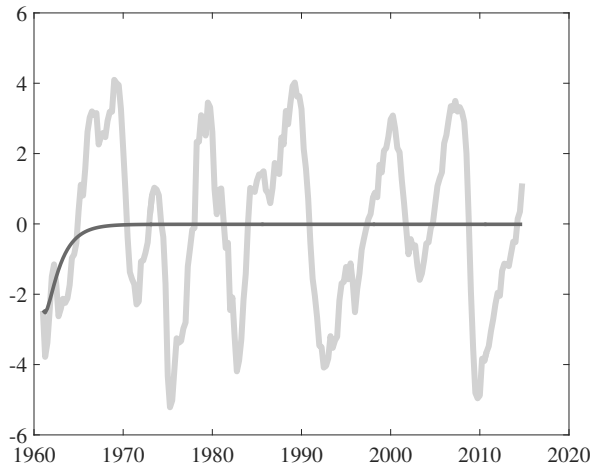
Figure 15: Dynamic Simulation, AR(2) Model



IV. Further Evidence

Common Way of Looking at the Data

Figure 16: Forecasting as of 1961, Linear Model



IV. Further Evidence

Full equation

$$\begin{cases} x_t &= \alpha_0 + \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \alpha_3 X_{t-1} + \alpha_4 x_{t-1}^3 + \varepsilon_t \\ X_t &= \sum_{j=0}^N (1 - \delta)^j x_{t-j} \end{cases}$$

IV. Further Evidence

Total Hours

Ordinary Least-squares Estimates

Dependent Variable = x

R-squared = 0.9424

Durbin-Watson = 2.1163

Nobs, Nvars = 220, 5

Variable	Coefficient	t-statistic	t-probability
constant	-0.020394	-0.526499	0.599084
x(-1)	1.391289	19.728102	0.000000
x(-2)	-0.342470	-5.121542	0.000001
X(-1)	-0.013430	-4.377008	0.000019
x(-1)^3	-0.006238	-2.386330	0.017884

Max Eigenvalue Modulus

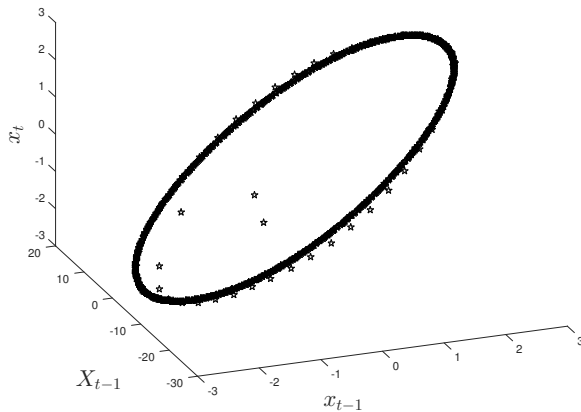
Full :1.0178 [1.01+0.12547i]

AR :0.85849 [0.85849]

IV. Further Evidence

Total Hours

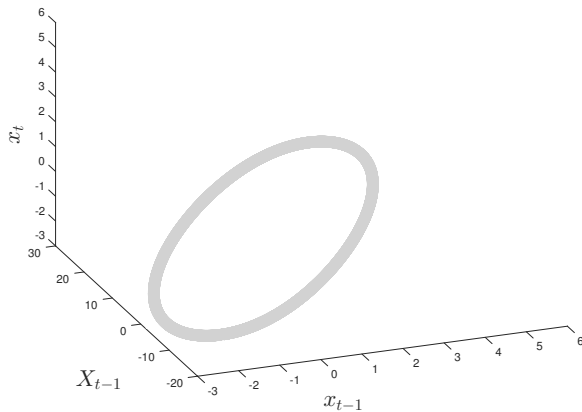
Figure 17: The Limit Cycle - Simulation as of $T_0 = 1961$



IV. Further Evidence

Total Hours

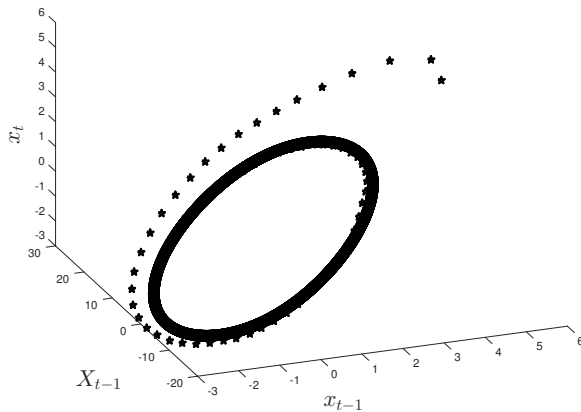
Figure 18: The Limit Cycle



IV. Further Evidence

Total Hours

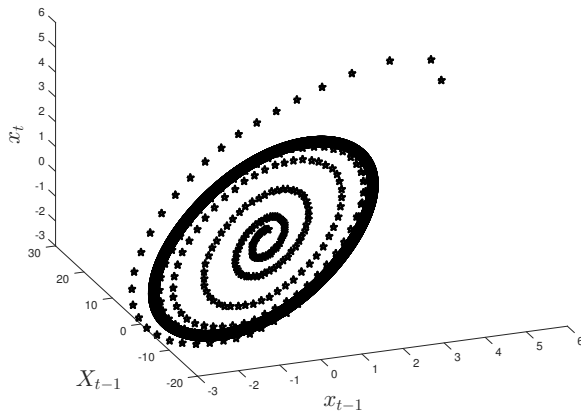
Figure 19: The Limit Cycle



IV. Further Evidence

Total Hours

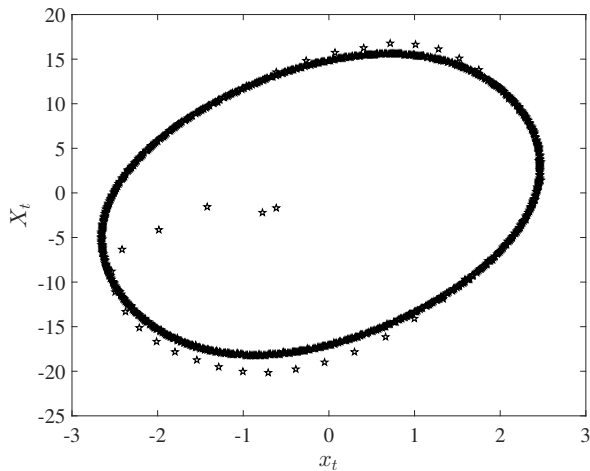
Figure 20: The Limit Cycle



IV. Further Evidence

Total Hours

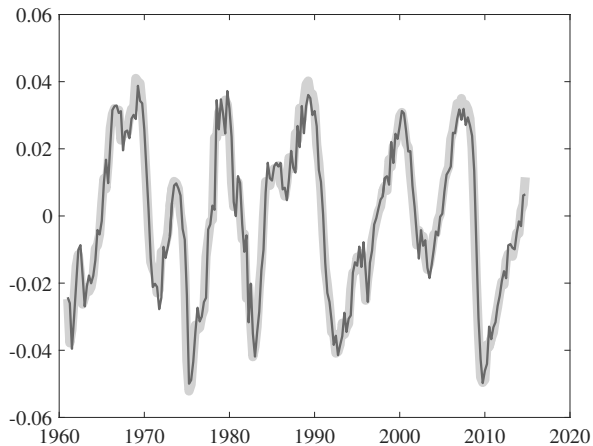
Figure 21: The Limit Cycle - Simulation as of $T_0 = 1961$



IV. Further Evidence

Total Hours

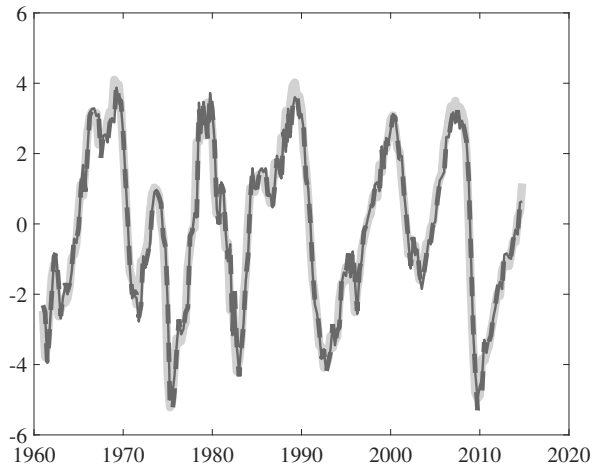
Figure 22: Dynamic Simulation, Non Linear Model



IV. Further Evidence

Total Hours

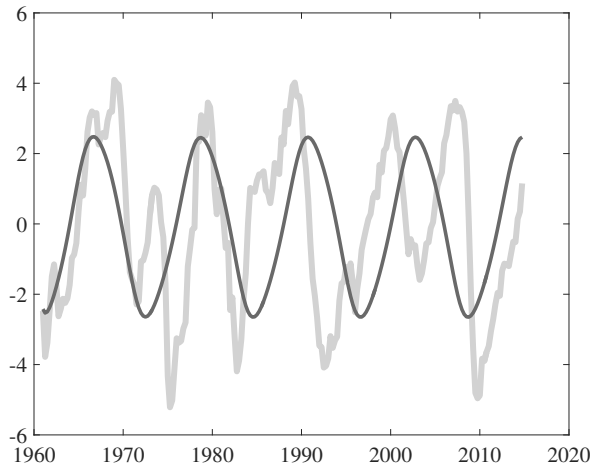
Figure 23: Dynamic Simulation, Full Model



IV. Further Evidence

Total Hours

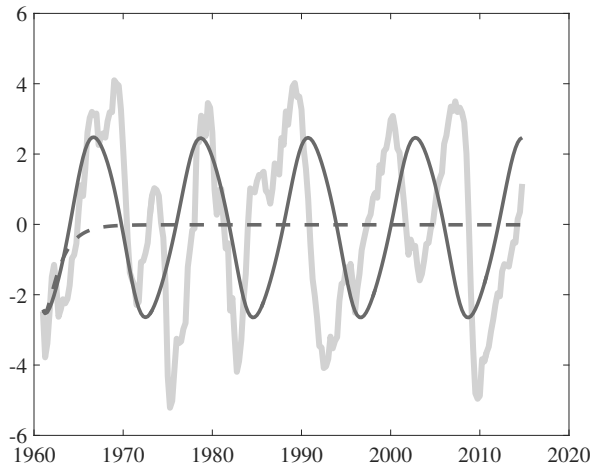
Figure 24: Forecasting as of 1961, Full Model



IV. Further Evidence

Total Hours

Figure 25: Forecasting as of 1961, AR(2) and Full Model



IV. Further Evidence

Total Hours

Figure 26: Visual Inspection of Non Linearities

$$x_t = \alpha_0 + \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \alpha_3 x_{t-1} + \alpha_4 x_{t-1}^3 + \varepsilon_t$$

